Computer Science

Theory of Computation

Undecidability

Lecture No.- 2



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Recap of Previous Lecture



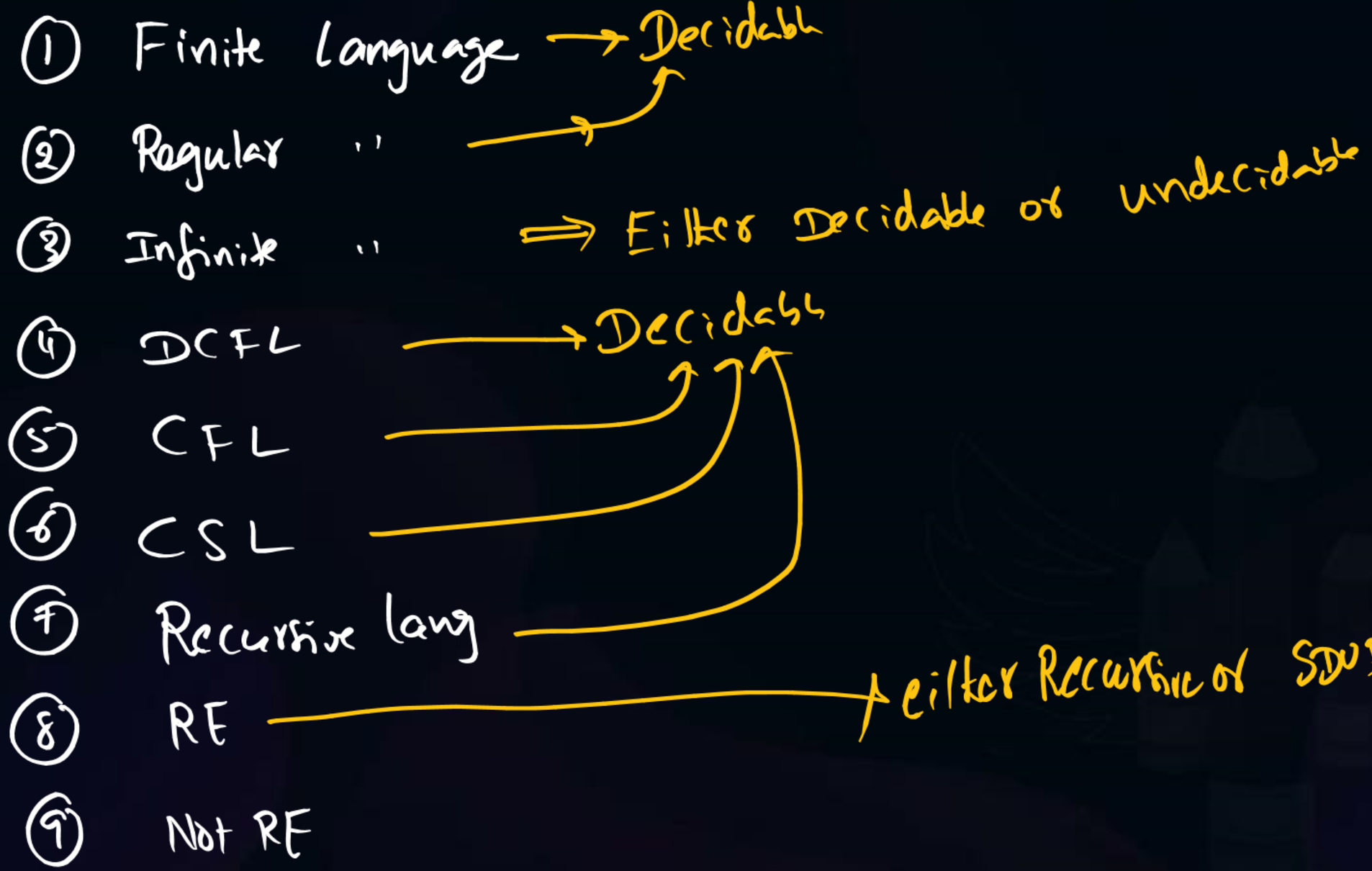
Topic Decision Properties ✓

Topics to be Covered

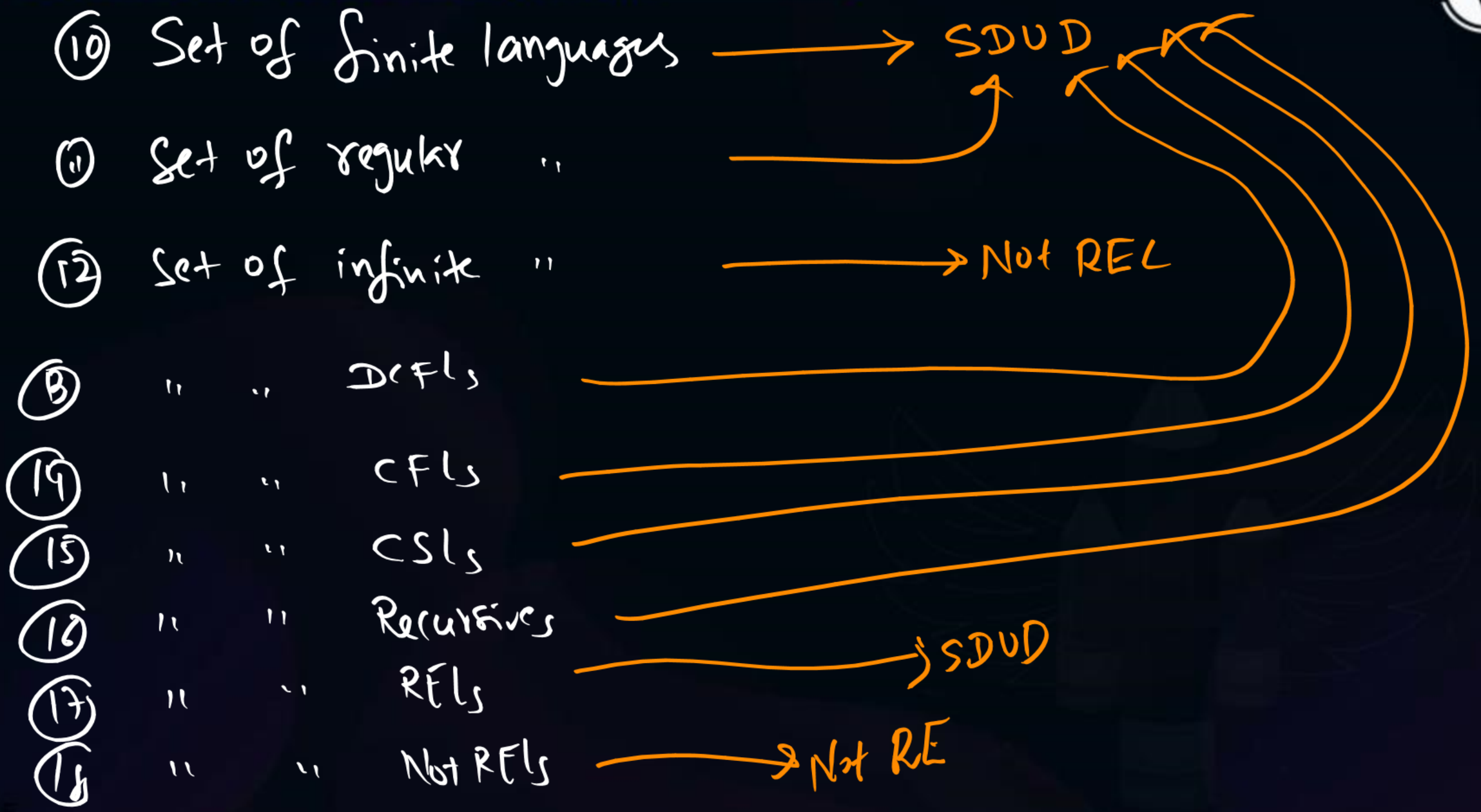


- Topic — Decidable
- Topic — SDUD
- Topic — RE
- Topic — Not RE
- Topic — Undecidable

Decidable and Undecidable problems



Decidable and Undecidable problems



Decidable and Undecidable problems

Finite Language

↳ Decidable

$= \{\varepsilon,$

$L = \{a^n b^n \mid n < 10\}$ finite set

↳ FA

$\bar{L}$ = Infinite

↳ FA

Set of finite languages

↳ SDUD

$L = \{F_1, F_2, F_3, \ldots\}$

↳ TM

$\ldots\ldots$

$\bar{L} = \{\bar{I}_1, \bar{I}_2, \bar{I}_3, \ldots\}$

↳ No TM

Decidable and Undecidable problems

(19) {TM | TM accepts ε} → SDUD

(20) {TM | TM accepts abc} → SDUD

IS TM accepts w?
↳ SDUD

(21) {TM | TM accepts nothing} → Not REL

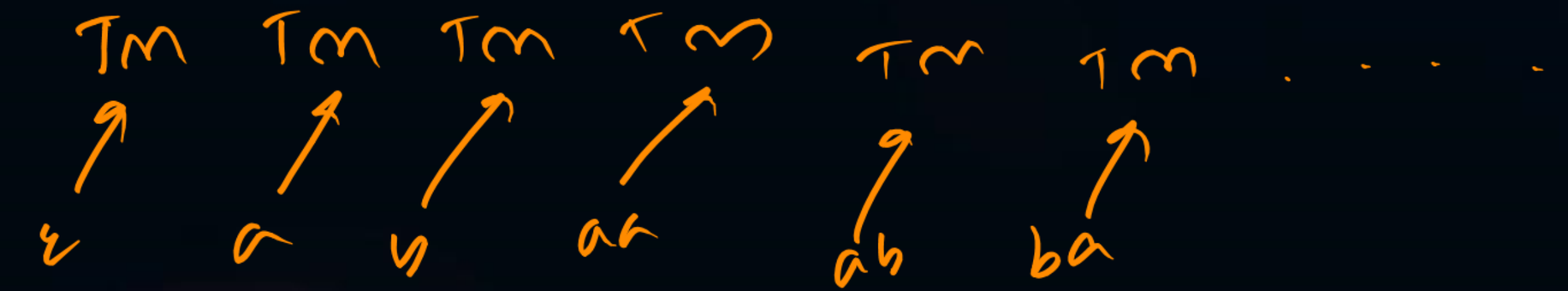
IS TM accepts ϕ?
IS L(TM) = ϕ? → Yes: X

(22) {TM | TM accepts Something} → SDUD

(23) {TM | TM accepts finite lang} → Not RE

(24) {TM | TM accepts everything} → Not RE

Decidable and Undecidable problems



If Tm accepts something, atleast one thread will halt

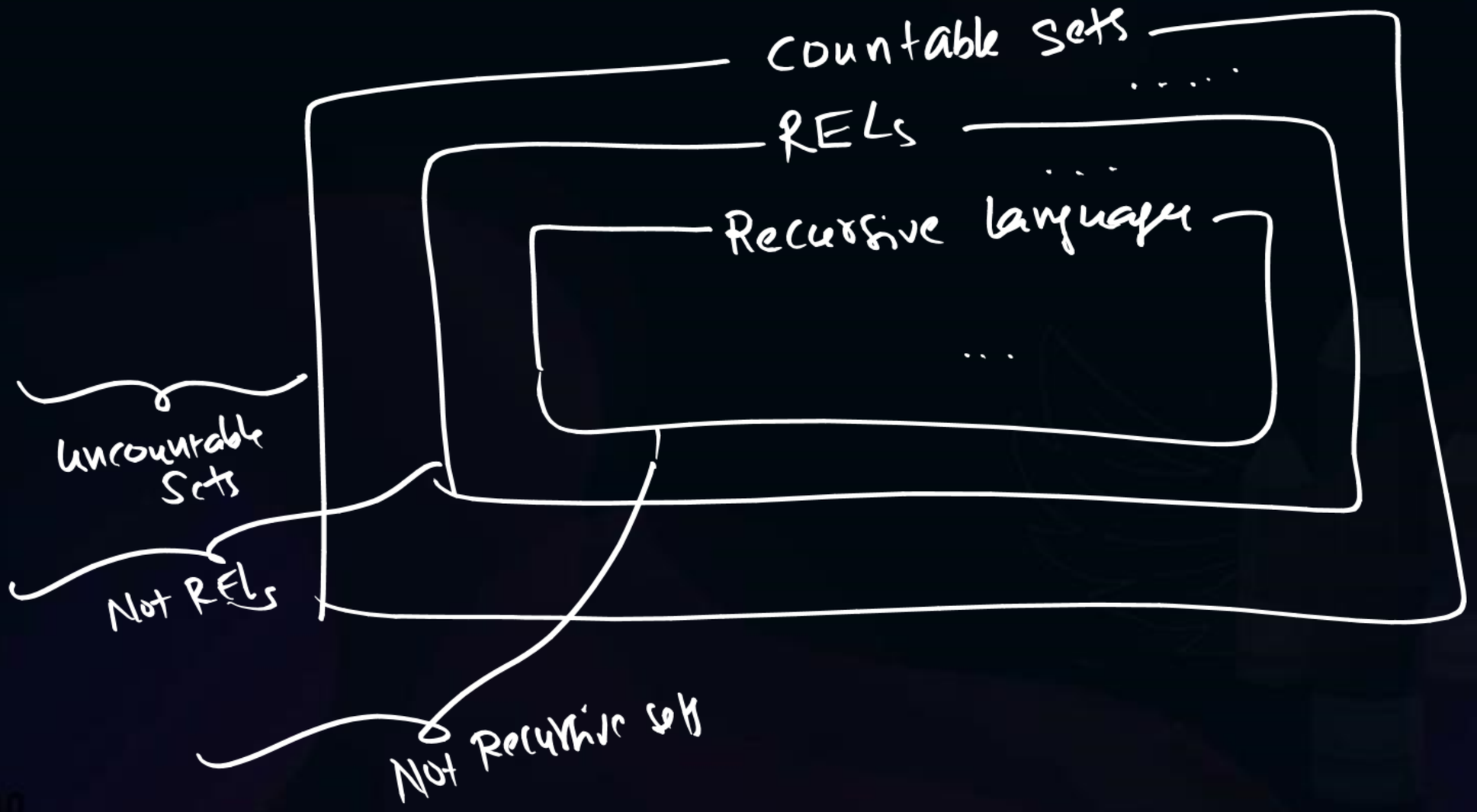
(25) $\{ \langle Tm \rangle \mid Tm \text{ accepts only } ab \} \longrightarrow$ Not RE

Is Tm accepts only ab ?

Is $L(Tm) = \{ab\}$

↳ Not RE

Decidable and Undecidable problems



Decidable and Undecidable problems

Set of all symbols = Σ → Finite set → Regular → Decidable, Countable

Set of all strings = Σ^* → Regular → Decidable; Infinite → Countable

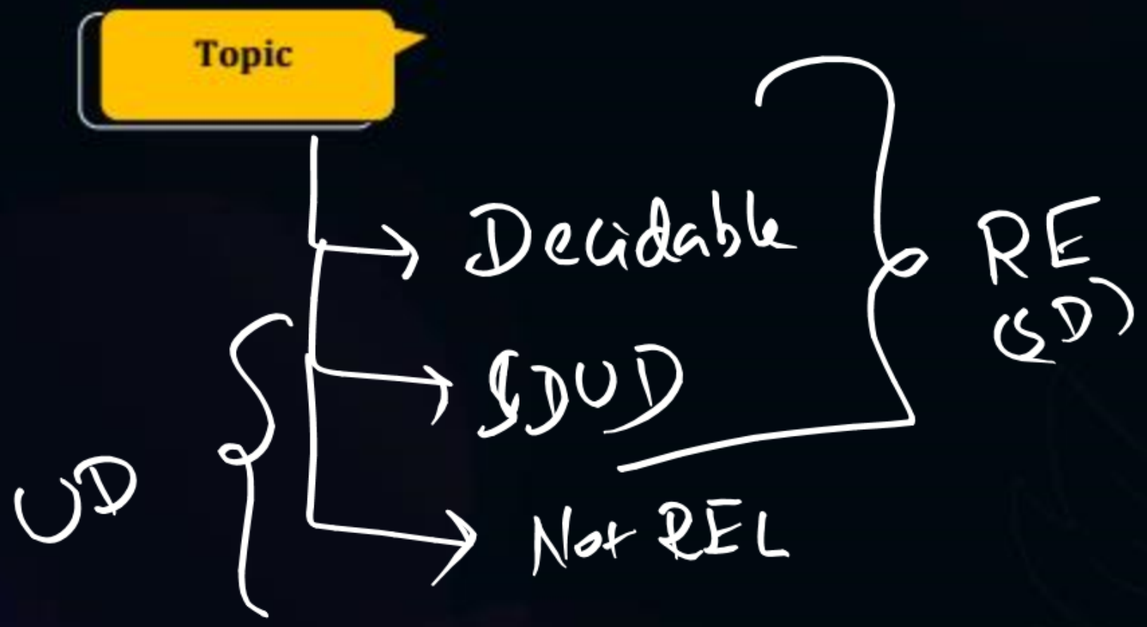
2^{Σ^*} = Powerset(Σ^*) = Set of all Sets

→ Not REL

→ Uncountable



2 mins Summary



THANK - YOU